

Software Requirements Specification (SRS) for plasmaERP

1. Introduction

1.1 Purpose

The purpose of this document is to define the software requirements for plasmaERP, a comprehensive Enterprise Resource Planning (ERP) system designed to manage and streamline various business processes including Human Resources, Finance, Sales, Inventory, Procurement, and Core Administration.

1.2 Scope

plasmaERP is a modular web-based application built using the Django framework. The system includes the following core modules:

- **Core & Accounts (Administration):** User, role, and permission management along with company and branch setup.
- **Human Resource Management (HRM):** Employee records, attendance, leave management, and payroll.
- **Finance & Accounting:** Chart of accounts, journal entries, expenses, and revenue management.
- **Sales:** Customer management, sales orders, invoices, and payments.
- **Inventory:** Product catalog, stock management, warehouses, and stock movements.
- **Procurement:** Supplier management, purchase orders, and goods receipts.
- **Approval & Audit:** System-wide approval workflows and audit logging for security and compliance.
- **Notifications:** Internal alert and notification system.

2. Overall Description

2.1 User Characteristics

The system will be used by various stakeholders within the organization:

- **Administrators:** Full access to all modules, configuration, user management, and audit logs.
- **HR Managers:** Access to HRM module for managing employees, attendance, leaves, and payroll.
- **Finance Officers:** Access to Finance module for managing ledgers, expenses, revenue, and payroll processing.
- **Sales Representatives:** Access to Sales module for handling customers, orders, and invoicing.
- **Inventory Managers:** Access to Inventory and Procurement modules for tracking stock and ordering supplies.
- **Standard Employees:** Limited access to view personal payroll, submit leave requests, and mark attendance.

2.2 Operating Environment

- **Server:** Linux/Unix server running Python (Django), PostgreSQL database.
- **Client:** Modern web browsers (Chrome, Firefox, Safari, Edge) on desktop and mobile devices.

3. System Features & Requirements

3.1 Accounts & Core Module

- **User Management:** System must allow creation, updating, and deactivation of user accounts linked to employees.
- **Role-Based Access Control (RBAC):** Define roles (e.g., Admin, HR, Sales) and assign specific permissions.
- **Company Setup:** Manage company details, branches, fiscal years, and cost centers.

3.2 Human Resource Management (HRM)

- **Employee Directory:** Maintain detailed employee profiles including designation, department, and branch.
- **Attendance Tracking:** Record daily check-in/check-out times and compute working hours.
- **Leave Management:** Employees can request leaves; managers can approve/reject them based on leave types.
- **Payroll Processing:** Generate monthly salary slips detailing basic salary, allowances, deductions, and net salary.

3.3 Finance Module

- **Chart of Accounts:** Hierarchical ledger system classifying assets, liabilities, equity, revenue, and expenses.
- **Journal Entries:** Double-entry bookkeeping system for manual adjustments.
- **Income & Expense Tracking:** Record business revenues and expenses against specific accounts.

3.4 Sales Module

- **Customer Management:** Maintain customer records, contact info, and credit limits.
- **Sales Orders:** Draft, confirm, and deliver sales orders.
- **Invoicing:** Generate invoices from sales orders, applying taxes and discounts.
- **Payments:** Record customer payments against specific invoices.

3.5 Inventory Module

- **Product Catalog:** Manage products with SKUs, barcodes, categories, and units of measurement.
- **Warehouse Management:** Track multiple warehouse locations.
- **Stock Tracking:** Monitor quantity, reserved quantity, and available quantity per product and warehouse.
- **Stock Movements:** Record stock IN, OUT, TRANSFER, and ADJUSTMENT operations.

3.6 Procurement Module

- **Supplier Management:** Maintain a database of suppliers and vendors.
- **Purchase Orders:** Create POs for suppliers, tracking expected delivery dates and total amounts.
- **Goods Receipts:** Record the receipt of items against POs into specific warehouses.

3.7 Cross-Functional Features

- **Approval Workflows:** Define multi-step approval processes for critical actions (e.g., Leave Approval, PO Confirmation).
- **Audit Logs:** Automatically track modifications (create, update, delete) to sensitive models.
- **Notifications:** Push internal system notifications for pending approvals, task assignments, and alerts.

- **Reporting:** Generate analytical reports across all modules (e.g., Sales Reports, Stock Valuation, Payroll Summary).

4. Non-Functional Requirements

4.1 Security

- Secure authentication system with password validation and session management.
- Data encryption in transit (HTTPS) and protection against common web vulnerabilities (CSRF, XSS, SQLi).

4.2 Performance

- System should load standard pages within 2 seconds.
- Database queries should be optimized to support concurrent users without locking.

4.3 Reliability & Availability

- System architecture must support 99.9% uptime during business hours.
- Automated daily backups of the PostgreSQL database.

5. Technology Stack

- **Backend Framework:** Django (Python)
- **Database:** PostgreSQL
- **Frontend:** HTML, CSS, JavaScript (Django Templates)
- **Architecture:** Monolithic architecture with modular Django apps.